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1. Abstract

Noetica: Mind and Relationship Analyser is a Python-based tool designed to enable structured self-reflection, cognitive evaluation, and relational insight. The platform addresses the limitations of unstructured introspection by integrating quantitative scoring with qualitative analysis to provide a nuanced understanding of mental processes and interpersonal behaviours. Its framework consists of three interconnected domains: Mind, Relationship, and Integrative Insight. The Mind domain evaluates self-awareness, emotional regulation, impulse control, and cognitive clarity. The Relationship domain examines empathy, communication, trust, and relational consistency. Integrative Insight explores the reciprocal influence of cognition and social interactions. User responses in a Yes/Maybe/No format are algorithmically scored and analysed to generate personalised feedback, actionable recommendations, and a seven-day reflective practice plan. Advanced visual analytics, including pie charts, histograms, bar plots, and line graphs, enhance interpretability. The tool exemplifies the convergence of computational methods, data visualisation, and psychological principles, offering an accessible, evidence-informed framework for deep self-understanding and sustained personal growth.

2. Introduction

Noetica: Mind and Relationship Analyser

The cultivation of self-awareness and the development of effective interpersonal skills are increasingly recognised as essential components of personal growth and overall well-being. In contemporary society, individuals are frequently exposed to complex social interactions, emotional challenges, and cognitive pressures that require both reflective thinking and adaptive relational skills. Despite the recognised importance of these competencies, many individuals lack structured mechanisms to evaluate their own thought patterns, emotional responses, and social behaviours systematically. This absence of structured self-assessment often leads to incomplete understanding of one's cognitive and relational tendencies, resulting in missed opportunities for personal development and growth.

Noetica: Mind and Relationship Analyser has been developed to address this need by providing a practical, interactive, and structured framework for self-reflection and assessment. The tool utilises the capabilities of Python programming to combine quantitative scoring with qualitative interpretation, transforming subjective self-reflection into measurable and actionable insights. By doing so, it enables users to explore their internal mental processes and social interactions in a systematic manner.

At the core of Noetica is a comprehensive questionnaire divided into three interrelated domains: Mind, Relationship, and Integrative Insight. The Mind domain evaluates cognitive and emotional processes, including self-awareness, emotional regulation, reflective thinking, impulse control, and clarity of thought. This domain allows users to gain insight into their internal cognitive patterns and the ways in which these patterns influence behaviour and decision-making. The Relationship domain examines interpersonal functioning, assessing aspects such as empathy, communication effectiveness, trust, boundary recognition, and consistency in social interactions. This domain provides users with an understanding of their relational dynamics and highlights areas where their interactions with others may be strengthened or improved. The Integrative Insight domain bridges the Mind and Relationship domains, emphasising the interplay between cognitive clarity and relational competence. It allows users to

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comprehend how their internal mental states influence their social behaviour and how relationships, in turn, affect their mental and emotional equilibrium.

User responses, captured in a Yes / Maybe / No format, are converted into numerical scores through a defined scoring methodology. The tool applies conditional logic and algorithmic evaluation to generate detailed, personalised feedback. This feedback identifies areas of strength, highlights aspects that may benefit from improvement, and provides practical, actionable guidance for enhancing both cognitive and relational skills. Additionally, Noetica offers a 7-day structured reflective practice plan, encouraging users to engage in daily introspection, mindful observation of thoughts and emotions, and incremental behavioural adjustments. This structured approach promotes ongoing self-awareness and fosters personal development over time.

To enhance user engagement and comprehension, Noetica integrates visual analytics, including pie charts to depict domain contributions, histograms to illustrate response frequencies, bar plots for comparative analysis of scores, and line graphs to reveal sequential patterns and trends across responses. The tool also generates a formal self-insight certificate, consolidating scores, analyses, and recommendations into a tangible artefact that users can retain for reflection or future reference.

In essence, Noetica demonstrates how Python programming, data analysis, and visualisation techniques can be applied to psychological self-assessment in a practical and meaningful way. By providing a structured, evidence-informed framework for self-reflection, the tool empowers users to enhance self-awareness, emotional understanding, and interpersonal competence, fostering holistic personal growth in an interactive and accessible format.

3. Problem Statement

In contemporary society, individuals often face challenges in understanding their own thoughts, emotions, and relational behaviours. Despite the abundance of self-help literature, reflective exercises, and psychological tools, many approaches remain unstructured, subjective, and inconsistent, making it difficult for individuals to obtain a clear and comprehensive understanding of their cognitive and relational patterns.

A significant issue arises from the gap between subjective self-reflection and objective assessment. Traditional self-reflection techniques rely heavily on personal interpretation and memory, which can introduce bias, inconsistency, and incomplete insights. Users may either overestimate their emotional regulation or fail to recognise recurring negative behavioural patterns, limiting the practical usefulness of unstructured reflection. Furthermore, existing tools that provide objective measurement often require professional intervention, clinical knowledge, or complex instruments, which restrict accessibility for individuals seeking personal growth in a simple, self-guided manner.

Another challenge lies in the integration of cognitive and relational self-assessment. Most tools focus exclusively on either internal mental processes or external social interactions, rarely examining how these two dimensions influence one another. In reality, cognitive clarity, emotional regulation, and interpersonal competencies are interdependent; the state of one often directly impacts the other. This lack of an integrative perspective prevents individuals from fully understanding the reciprocal relationship between their thoughts, emotions, and social interactions, and limits opportunities for targeted personal development.

Additionally, users frequently lack actionable guidance following self-assessment. While many questionnaires or reflective exercises provide scores or descriptive summaries, they often fail to translate these insights into concrete steps for improvement. Without structured guidance, individuals may struggle to convert

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awareness into behaviour modification, reducing the overall efficacy of self-reflection practices.

The problem can therefore be summarised as follows:

1. **Need for Structured Self-Reflection:** Individuals require a systematic and consistent approach to evaluate their mental and relational patterns accurately.
2. **Accessibility of Objective Assessment:** There is a demand for user-friendly tools that can objectively assess cognitive and relational aspects without the need for professional intervention.
3. **Holistic Integration of Mind and Relationships:** Users benefit from tools that examine the interplay between internal cognition and external social dynamics, providing a comprehensive understanding of personal development.
4. **Actionable Feedback and Growth Facilitation:** Effective tools should provide personalised, evidence-informed recommendations, empowering users to implement incremental and sustainable improvements in thought patterns, emotional regulation, and relational behaviours.

Noetica: Mind and Relationship Analyser addresses these challenges by combining structured questionnaires, a numerical scoring system, conditional logic analysis, visualisation techniques, and a guided reflective practice plan. By bridging the gap between introspection and objective evaluation, the tool provides an accessible, integrative, and actionable framework for personal growth, enabling users to gain deeper awareness of themselves and their social interactions while fostering sustained cognitive, emotional, and relational development.

4. Project Overview

Noetica: Mind and Relationship Analyser is a Python-based mini-project designed to provide users with a structured and comprehensive framework for self-reflection, cognitive assessment, and relational insight. By integrating quantitative scoring with qualitative feedback, the project allows users to systematically explore their thought patterns, emotional regulation, and interpersonal behaviours, facilitating a holistic understanding of personal development.

This section presents a detailed overview of the project's purpose, scope, and methodology, as well as its intended objectives and goals. It serves to contextualise the implementation and outcomes of the tool, emphasising both its practical application and its theoretical grounding in psychological principles.

4.1 Objectives

Noetica is designed as a structured framework for self-reflection, cognitive assessment, and relational insight. Its purpose is to provide users with an accessible, interactive tool that transforms subjective introspection into measurable and actionable outcomes. The project utilises a three-domain evaluation system, Mind, Relationship, and Integrative Insight, to ensure a holistic understanding of personal development. Users respond to targeted questions in a Yes / Maybe / No format, which are converted into numerical scores to enable both quantitative assessment and qualitative reflection.

The primary objectives of Noetica are as follows:

1. **Enhance Self-Awareness:** Assess key aspects of cognitive and emotional functioning, including self-awareness, impulse control, emotional regulation, reflective capacity, and clarity of thought. This enables users to identify recurring patterns in their thoughts and behaviours, understand their internal decision-making mechanisms, and develop insight into personal cognitive strengths and limitations. It also supports deliberate

self-examination, promoting mindfulness and intentional observation of internal mental states.

2. **Promote Relational Insight:** Evaluate interpersonal competencies such as empathy, communication effectiveness, trust, boundary recognition, and relational consistency. This allows users to gain a clear understanding of their social interactions, highlights strengths, and identifies behaviours that may hinder or improve relationships. The tool provides guidance to enhance relational skills and foster healthier, balanced, and mutually respectful interactions.
3. **Facilitate Integrative Understanding:** Emphasise the interconnectedness of cognitive clarity, emotional regulation, and relational competence. Noetica demonstrates how internal mental states influence social interactions and how relational experiences impact emotional and cognitive functioning. Users are encouraged to adopt a holistic perspective on personal growth, recognising the reciprocal effects of mind and relationships.
4. **Provide Actionable Feedback:** Convert quantitative scores into personalised, evidence-informed recommendations. Feedback highlights areas of strength and improvement and provides practical strategies for enhancing cognitive, emotional, and relational capacities. Users are supported in translating self-awareness into measurable, incremental behaviour changes.
5. **Enhance Engagement through Visualisation:** Utilise visual analytics, including pie charts, histograms, bar plots, and line graphs, to present results in an intuitive and interpretable format. Visual tools allow users to recognise trends, domain imbalances, and sequential patterns in thought and relational behaviours, promoting active engagement with self-reflective data.
6. **Demonstrate Practical Application of Computational Tools:** Showcase the use of Python programming, data handling, conditional logic, and visualisation in psychological self-assessment. Noetica demonstrates how computational methods can support structured self-reflection, provide actionable insights, and facilitate personal development, even in a mini-project context.

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By achieving these objectives, Noetica provides users with a systematic, integrative, and interactive framework for exploring their cognitive patterns, emotional regulation, and interpersonal behaviours, fostering holistic personal growth.

4.2 Goals

The overarching goal of Noetica is to deliver an interactive, evidence-informed, and structured framework for self-assessment that bridges the gap between subjective introspection and objective evaluation. The project aims to empower individuals to understand themselves more deeply, enhance relational competencies, and implement actionable personal growth strategies.

The specific goals of Noetica include:

1. **Structured Self-Reflection:** Encourage deliberate and consistent self-examination of thoughts, emotions, and behaviours. Help users identify habitual cognitive patterns, emotional responses, and behavioural tendencies. Promote ongoing self-awareness and reflective practice, enabling better understanding of internal mental processes.
2. **Relational Competence Enhancement:** Support evaluation and improvement of interpersonal skills such as empathy, effective communication, trustworthiness, and boundary-setting. Highlight both strengths and areas requiring improvement in social interactions. Facilitate the development of healthier and more fulfilling relationships.
3. **Holistic Integration of Mind and Relationships:** Provide insight into the dynamic interplay between cognitive clarity, emotional regulation, and relational functioning. Help users understand how internal mental states affect social interactions and how relationships influence emotional and cognitive well-being. Encourage a comprehensive approach to personal development that considers both intrapersonal and interpersonal dimensions.
4. **Actionable Insights and Guided Development:** Convert assessment data into personalised recommendations and a structured seven-day reflective

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practice plan. Enable users to implement incremental improvements in cognitive, emotional, and relational capacities. Bridge awareness and action by translating insights into practical strategies for sustained personal growth.

5. Engagement through Intuitive Visualisation: Present results in a visually engaging and interpretable format using pie charts, histograms, bar plots, and line graphs. Allow users to identify domain imbalances, recurring patterns, and trends in their responses to support deeper self-reflection. Support data-driven personal development while maintaining simplicity and accessibility for all users.
6. Showcase Practical Application of Python and Analytical Tools: Demonstrate how Python programming, data analysis, conditional logic, and visualisation can be applied to psychological self-assessment. Show how computational tools can transform subjective introspection into structured, actionable, and measurable outcomes.

Through these goals, Noetica provides a comprehensive, interactive, and evidence-informed tool that facilitates self-understanding, emotional regulation, relational improvement, and sustained personal development. It highlights the potential of combining computational methods with psychological principles to create meaningful and practical frameworks for holistic self-growth.

5. Implementation

The implementation phase of Noetica: Mind and Relationship Analyser details the practical development of the tool, illustrating how the project objectives, goals, and design principles are translated into a functional Python-based system. This section describes the problem analysis, methodology, modular structure, and coding practices employed to create an interactive, structured, and insightful self-assessment platform.

5.1 Problem Analysis and Description

The primary problem addressed by Noetica is the absence of a structured, accessible, and integrative tool that enables users to systematically reflect on both their internal cognitive processes and external relational behaviours. Many existing self-reflection exercises lack quantifiable assessment, personalised feedback, or clear guidance for improvement. Conversely, traditional psychological assessment tools may be overly complex, clinically oriented, or inaccessible without professional guidance.

The problem can be described as follows:

1. **Subjective Self-Reflection Limitations:** Individuals often engage in unstructured introspection, which may result in partial, biased, or inconsistent self-perception. There is a need for a systematic approach to guide reflection, track responses, and identify behavioural or cognitive patterns over time.
2. **Integration of Cognitive and Relational Assessment:** Existing tools often assess cognition or relational skills separately. A central challenge is to develop an integrative system that considers how cognitive clarity, emotional regulation, and relational competence influence one another, providing a more holistic understanding of personal development.

3. **Actionable Feedback and Engagement:** Users benefit most when insights are not only descriptive but also actionable. Therefore, a solution must translate scores into meaningful feedback, visualise patterns for intuitive comprehension, and offer structured guidance, such as a daily reflective plan, to encourage incremental improvement.

In addressing these challenges, Noetica provides a Python-based interactive platform that combines structured assessment, algorithmic analysis, visualisation, and guided reflection, enabling users to explore, understand, and enhance both their internal and external personal competencies.

5.2 Methodology / Algorithms Used

The methodology for implementing Noetica involves systematic data collection, scoring, analysis, and feedback generation, facilitated by Python programming and data visualisation libraries. Key steps include:

1. **Structured Questionnaire Design:** The tool employs a comprehensive set of questions, divided into three domains: Mind, Relationship, and Integrative Insight. Each question is designed to probe self-awareness, cognitive functioning, emotional regulation, relational skills, or the interconnection between mind and relationships.
2. **Scoring Algorithm:** User responses, collected in a Yes / Maybe / No format, are converted into numerical scores using a predefined mapping: Yes = 2, Maybe = 1, No = 0. This scoring system allows for quantitative evaluation while retaining the simplicity and accessibility necessary for a mini-project context.
3. **Conditional Logic Analysis:** Conditional statements are applied to total scores in each domain to generate personalised textual feedback. This feedback highlights strengths, identifies areas for improvement, and provides actionable guidance tailored to the user's profile.
4. **Visualisation Techniques:** Python's matplotlib and seaborn libraries are used to generate multiple types of visual representations:

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- a. Pie Charts: Show proportional contributions of Mind and Relationship domains.
 - b. Histograms: Illustrate the frequency of responses in each scoring category.
 - c. Bar Plots: Compare scores across domains to identify relative strengths and weaknesses.
 - d. Line Graphs: Display sequential response patterns and trends, supporting reflection on consistency and variation.
5. Structured Reflective Plan Algorithm: Based on the analysis, the tool generates a seven-day reflective plan using if-then logic. Users are guided to observe thoughts, emotions, and relational patterns incrementally, fostering daily awareness and gradual behavioural refinement.

The methodology emphasises simplicity, accessibility, and interpretability, ensuring that even users without prior technical or psychological training can engage meaningfully with the tool.

5.3 Modules Identified

The implementation of Noetica is modular, facilitating clarity, maintainability, and stepwise execution. The major modules identified are:

1. User Interface and Input Module
 - a. Handles user interaction through the console, prompting for personal details, age group, gender, and responses to questionnaire items.
 - b. Validates input and ensures consistent response formats (Yes / Maybe / No).
2. Scoring and Analysis Module
 - a. Converts responses into numerical scores.

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- b. Aggregates scores by domain (Mind, Relationship, Integrative Insight).
 - c. Applies conditional logic to generate textual feedback, highlighting strengths, weaknesses, and recommendations.
- 3. Visualisation Module
 - a. Generates graphical representations of results, including pie charts, histograms, bar plots, and line graphs.
 - b. Enhances interpretability and engagement by allowing users to observe trends, distributions, and domain comparisons.
- 4. Reflective Practice Module
 - a. Constructs a seven-day structured reflective plan tailored to the user's profile.
 - b. Encourages incremental observation, self-reflection, and behavioural refinement.
- 5. Certificate Generation Module
 - a. Compiles user details, scores, feedback, and recommendations into a formal self-insight certificate.
 - b. Provides a tangible artifact for personal reflection and future comparison.

Each module is interconnected yet independently functional, enabling smooth execution and simplifying potential updates or extensions to the tool.

5.4 Code with Comments

This section presents the key implementation components of the *Noetica: Mind and Relationship Analyser*, supported by inline comments within the Python code to ensure clarity, readability, and ease of understanding. The purpose of including the commented code is to demonstrate how theoretical concepts such as scoring, comparison, and interpretation are translated into functional program logic.

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The code is written in Python and executed in a Google Colab environment. Each major functional block is clearly labelled using comments, allowing readers to identify the role of individual sections without requiring prior in-depth programming expertise.

Visualisation Module Implementation

Listing 5.4 presents the complete implementation of the analyser, including the visualisation module responsible for generating graphical representations of the analysed psychological data. Rather than illustrating a single isolated visual element, the listing demonstrates how multiple visual outputs are produced as part of the overall analytical workflow.

Within the visualisation module, graphical representations such as pie charts, histograms, bar plots, and line plots are generated to convey different aspects of the analysed data. These visualisations collectively display proportional balance between the Mind and Relationship domains, response frequency distributions, comparative domain scores, and response flow patterns.

The matplotlib and seaborn libraries are used to initialise figures with fixed dimensions to ensure visual balance and readability. Aggregated scores derived from the Mind and Relationship assessments are utilised as inputs for the visualisation functions, with labels, legends, and titles applied to enhance clarity and interpretability. Consistent colour schemes and layout parameters are used across all visual outputs to maintain coherence.

Inline comments, denoted by the `#` symbol, are used throughout the implementation to explain the intent of each operation rather than merely restating syntax. This approach ensures that the code communicates both functionality and underlying reasoning, which is essential for academic evaluation, reproducibility, and future maintainability.

```
# Mind & Relationship Analyser

"""
Formal Self-Reflection & Insight Tool

This platform provides structured psychological self-awareness based on
self-reported responses. It is intended for reflection and education,
NOT diagnosis or clinical evaluation.
"""

# Imports

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from datetime import datetime

sns.set_style("whitegrid")

# Formal Introduction

print("\n=====")

print("MIND & RELATIONSHIP SELF-INSIGHT REPORT")
```

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```
print("=====")

print(
    "This report explores current mental and relational patterns based\n"
    "on\n"
    "self-reported responses. It reflects tendencies, not fixed\n"
    "traits,\n"
    "and is intended for awareness and personal growth.\n"
)

# Personal Information

print("Personal Information\n")

name = input("Name / Preferred Name: ").strip()

print("\nAge Group:")

print("1. Under 18")

print("2. 18-25")

print("3. 26-35")

print("4. 36+")

age_map = {
    "1": "Under 18",
    "2": "18-25",
    "3": "26-35",
```

```
        "4": "36+"
    }

age_group = age_map.get(input("Choice (1-4): ").strip(), "Not
specified")

print("\nGender (optional):")

print("1. Male")

print("2. Female")

print("3. Non-binary")

print("4. Prefer not to say")

gender_map = {

    "1": "Male",

    "2": "Female",

    "3": "Non-binary",

    "4": "Prefer not to say"

}

gender = gender_map.get(input("Choice (1-4): ").strip(), "Prefer not to
say")

# Local date + time

date_time = datetime.now().astimezone().strftime("%d %B %Y, %I:%M %p")

# Response Format
```

```
print("\nResponse format:")

print("y / Y / yes / Yes / YES = Yes")

print("m / M / maybe / Maybe / MAYBE = Maybe")

print("n / N / no / No / NO = No\n")


accepted_inputs = {

    "y": "Yes", "Y": "Yes", "yes": "Yes", "Yes": "Yes", "YES": "Yes",

    "m": "Maybe", "M": "Maybe", "maybe": "Maybe", "Maybe": "Maybe",

    "MAYBE": "Maybe",

    "n": "No", "N": "No", "no": "No", "No": "No", "NO": "No"

}


score_map = {"Yes": 2, "Maybe": 1, "No": 0}


def get_response(question):

    while True:

        ans = input(f"{question} (Y/M/N): ").strip()

        if ans in accepted_inputs:

            return score_map[accepted_inputs[ans]]

        print("Invalid input. Please use the defined response formats.")


# Questions
```

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```
mind_questions = [  
    "Do I clearly know my own flaws?",  
    "Can I stay calm when my emotions rise?",  
    "Can I admit I'm wrong without defending myself?",  
    "Am I open to changing my beliefs when proven wrong?",  
    "Do I take responsibility for my life's outcomes?",  
    "Does my life feel meaningful to me right now?",  
    "Do my actions usually match my values?",  
    "Can I control my impulses most of the time?",  
    "Do I face fear and uncertainty instead of escaping it?",  
    "Do I regularly reflect on my thoughts and behavior?",  
    "Can I quiet my mind when I need to rest or focus?",  
    "Do small problems affect my mood longer than they should?",  
    "Do I notice when my thinking becomes unhealthy or unbalanced?",  
    "Do I repeat the same mistakes despite understanding them?",  
    "Do I feel mentally balanced when under pressure?",  
    "Can I focus on the present instead of past or future?",  
    "Do I notice bodily sensations before emotional reactions?",  
    "Can I maintain calm when my mind is cluttered?",  
    "Do I have clarity about what truly matters to me?",  
    "Do I intentionally guide my thoughts instead of reacting  
automatically?"  
]  
  
relationship_questions = [
```

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```
"Do I truly feel and understand others' emotions?",  
"Am I honest with people even when it's uncomfortable?",  
"Do I respect others' boundaries without resentment?",  
"Do I treat everyone with basic dignity, regardless of status?",  
"In conflict, do I aim to resolve rather than win?",  
"Can people rely on my words and promises?",  
"When I hurt someone, do I accept responsibility fully?",  
"Do I love without trying to control or possess?",  
"Do I hold myself to the same standards I expect from others?",  
"Am I fully present when someone is talking to me?",  
"Do I feel emotionally safe expressing myself in close  
relationships?",  
"Can I express my needs without guilt or fear?",  
"Can I stay close to people without losing myself?",  
"Do I process conflicts instead of avoiding them?",  
"Do I feel valued without having to prove my worth?",  
"Do I practice empathy even when it is hard?",  
"Do I allow others to be themselves without judgment?",  
"Do I notice when my actions hurt someone and correct it?",  
"Do I maintain warmth and connection even under stress?",  
"Do I actively nurture important relationships?"
```

```
]
```

```
integrative_question = "Do your relationships significantly influence  
your mental clarity and emotional balance?"
```

```
# Collect Responses

print("\n--- Mind Section ---\n")

mind_scores = [get_response(q) for q in mind_questions]

print("\n--- Relationship Section ---\n")

relationship_scores = [get_response(q) for q in relationship_questions]

print("\n--- Integrative Reflection ---\n")

integrative_score = get_response(integrative_question)

mind_total = sum(mind_scores)

relationship_total = sum(relationship_scores)

# Data for Bar Plot (FIX B.1)

df_bar = pd.DataFrame({

    "Category": ["Mind", "Relationship"],

    "Score": [mind_total, relationship_total]

})

# Analyses

def mind_analysis(score):
```

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```
    if score <= 12:

        return "You may feel mentally scattered or weighed down by your
thoughts. Life might feel overwhelming at times, and it's easy to get
stuck in loops of worry or self-doubt."

    elif score <= 24:

        return "You have moments of clarity and calm, but stress and
distractions still affect you. You sometimes notice your patterns, yet
old habits sneak in."

    elif score <= 32:

        return "Your mind is generally steady. You can reflect on your
thoughts, manage emotions, and stay grounded most of the time, though
occasional stress shows up."

    elif score <= 36:

        return "You have strong mental control and awareness. You
notice your inner world clearly, manage your impulses, and remain
present even under pressure."

    else:

        return "You are exceptionally in tune with your mind. Thoughts,
emotions, and focus are under your guidance, giving you clarity,
balance, and deep self-awareness."

def relationship_analysis(score):

    if score <= 12:

        return "Relationships may feel challenging or draining. You
might struggle with trust, setting boundaries, or being fully open with
others."

    elif score <= 24:

        return "You care deeply about connections, but sometimes
emotions or misunderstandings create tension. You're aware, but growth
is ongoing."

    elif score <= 32:
```

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```
        return "You are capable of healthy, supportive relationships.
You listen, respect boundaries, and maintain closeness while honoring
yourself."

    elif score <= 36:

        return "Your relationships are stable and nurturing. You
understand and connect with others while staying emotionally grounded."

    else:

        return "You experience secure, deep, and balanced
relationships. You love authentically, communicate clearly, and create
a safe space for others and yourself."

print("\nMind Analysis:")

print("-", mind_analysis(mind_total))

print("\nRelationship Analysis:")

print("-", relationship_analysis(relationship_total))

print("\nIntegrative Insight:")

if integrative_score == 0:

    print("- Your mind and relationships feel separate. What happens
internally may not influence your connections, and vice versa.")

elif integrative_score == 1:

    print("- Your inner world and your relationships sometimes affect
each other. You notice patterns, but not always consistently.")

else:

    print("- Your mind and relationships are deeply connected. How you
think and feel profoundly shapes your connections, and your connections
influence your inner balance.")
```

```
# Suggestions

def mind_suggestions(score):

    if score <= 12:

        return [

            "Take small breaks to clear your mind when overwhelmed.",

            "Notice when thoughts loop and gently redirect them.",

            "Write down or reflect on recurring worries to understand them.",

            "Practice calming routines like breathing or short walks.",

            "Be gentle with yourself; mental clarity grows gradually."

        ]

    elif score <= 32:

        return [

            "Observe your thoughts and emotions regularly.",

            "Practice focusing on what matters most right now.",

            "Notice patterns without judgment and learn from them.",

            "Take time for reflective or creative activities.",

            "Check in with yourself during stressful moments."

        ]

    else:

        return [

            "Keep cultivating awareness to stay balanced.",

            "Notice subtle changes in mood or focus.",
```

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```
        "Share your clarity to help others without overextending.",
        "Use mindfulness and reflection to deepen insight.",
        "Continue practicing presence and intentional thinking."
    ]

def relationship_suggestions(score):
    if score <= 12:
        return [
            "Set small boundaries to protect your energy.",
            "Notice where you feel drained or over-responsible.",
            "Practice listening without needing to fix everything.",
            "Be honest gently, even if it feels uncomfortable.",
            "Celebrate small moments of connection without pressure."
        ]
    elif score <= 32:
        return [
            "Share feelings clearly and listen fully.",
            "Balance closeness with personal space.",
            "Work through small conflicts before they grow.",
            "Notice what makes relationships feel safe or tense.",
            "Appreciate the people who support you and be present for
them."
        ]
    else:
        return [
```

```
        "Maintain openness and honesty while protecting yourself.",

        "Encourage growth in others without losing your own
balance.",

        "Keep communication clear and compassionate.",

        "Celebrate healthy boundaries and mutual trust.",

        "Deepen meaningful connection while honoring
individuality."

    ]

print("\nMind Suggestions:")

for s in mind_suggestions(mind_total):

    print("-", s)

print("\nRelationship Suggestions:")

for s in relationship_suggestions(relationship_total):

    print("-", s)

# Total Score Quote

total_score = mind_total + relationship_total + integrative_score #
0-82

total_score_quotes = {

    (0, 11): ("Happiness is the meaning and the purpose of life, the
whole aim and end of human existence.", "Aristotle"),

    (12, 23): ("Knowing yourself is the beginning of all wisdom.",
"Anaximander"),
```

```
(24, 35): ("Happiness depends upon ourselves.", "Gautam Buddha"),

(36, 47): ("He who conquers himself is the mightiest warrior.",
"Mozi"),

(48, 59): ("It is not what we have, but what we enjoy, that
constitutes our abundance.", "Pacal"),

(60, 71): ("Do not spoil what you have by desiring what you have
not.", "Al-Farabi"),

(72, 82): ("Peace comes from within; do not seek it without.",
"Tupaia")
}

def get_total_quote(total):

    for score_range, (quote, author) in total_score_quotes.items():

        if score_range[0] <= total <= score_range[1]:

            return quote, author

    return "Keep observing yourself and your path.", "Unknown"

quote, author = get_total_quote(total_score)

# 7-DAY PRACTICE (RETEST AFTER 7 DAYS)

print("\n=====")

print("7-DAY PRACTICE (RETAKE THIS TEST AFTER 7 DAYS)")

print("=====")

def seven_day_task(mind, rel):
```

```
combined = mind + rel

if combined <= 24:

    return (

        "For the next 7 days:\n"

        "- Each night, write ONE honest sentence about what  
disturbed your mind or relationships that day.\n"

        "- Do not judge it. Do not fix it. Just see it  
clearly.\n\n"

        "This builds awareness before change."

    )

elif combined <= 48:

    return (

        "For the next 7 days:\n"

        "- Each day, pause once when emotions rise and ask:\n"

        "  'What am I reacting to right now?'\n"

        "- Respond a little slower than usual.\n\n"

        "This weakens automatic reactions."

    )

elif combined <= 64:

    return (

        "For the next 7 days:\n"

        "- Choose ONE recurring habit or reaction.\n"

        "- Watch it carefully every time it appears.\n"

        "- Do not suppress it. Do not indulge it.\n\n"

        "This strengthens conscious control."
```

```
)

else:

    return (

        "For the next 7 days:\n"

        "- Maintain awareness of thoughts, emotions, and actions\n"
        "together.\n"

        "- Act deliberately, not automatically.\n"

        "- Speak only what feels true and necessary.\n\n"

        "This deepens clarity and balance."

    )

print(seven_day_task(mind_total, relationship_total))

print("\nAfter 7 days, take this test again and compare your scores.")

print("Growth is measured by awareness, not perfection.")

print("=====")

# Certificate

print("\n=====")

print("SELF-INSIGHT CERTIFICATE")

print("=====")

print(f"Name           : {name}")

print(f"Age Group        : {age_group}")

print(f"Gender           : {gender}")
```

Noetica: Mind and Relationship Analyser

```
print(f>Date & Time          : {date_time}")

print("-----")

print(f>Mind Score           : {mind_total} / 40")

print(f>Relationship Score : {relationship_total} / 40")

print(f>Integrative Score  : {integrative_score} / 2")

print(f>Total Score        : {total_score} / 82\n")

print("Conclusion:")

if mind_total >= 36 and relationship_total >= 36 and integrative_score
== 2:

    print("- You are in a strong place mentally and relationally. You
understand yourself, manage your emotions, and connect deeply with
others.")

elif mind_total <= 12 and relationship_total <= 12 and
integrative_score <= 1:

    print("- You may be experiencing internal and relational strain.
Awareness and small steps toward self-care and honest connection can
make a big difference.")

else:

    print("- You have areas of strength and areas to grow. Paying
attention to your mind and your relationships will help create balance
and deeper fulfillment.")

print("\nReflection Quote:")

print(f>"{quote}" - {author}")

print("\nComment:")
```

```
print("This certificate reflects current psychological patterns, not  
fixed traits. Awareness is the first step toward meaningful change.")  
  
print("=====  
  
# VISUALISATIONS  
  
# Pie Chart  
  
plt.figure(figsize=(5,5))  
  
plt.pie(  
    [mind_total, relationship_total],  
    labels=["Mind", "Relationship"],  
    autopct="%1.1f%%",  
    colors=["#4A90E2", "#7ED321"],  
    startangle=90  
)  
  
plt.title("Psychological Balance Distribution")  
  
plt.show()  
  
# Histogram  
  
plt.figure(figsize=(6,4))  
  
bins = [-0.5, 0.5, 1.5, 2.5]  
  
plt.hist(  
    mind_scores,  
    bins=bins,
```

```
        alpha=0.6,

        label="Mind",

        rwidth=0.9

    )

plt.hist(

    relationship_scores,

    bins=bins,

    alpha=0.6,

    label="Relationship",

    rwidth=0.9

)

plt.title("Score Frequency Distribution")

plt.xlabel("Response Score (0 = No, 1 = Maybe, 2 = Yes)")

plt.ylabel("Frequency")

plt.legend()

plt.tight_layout()

plt.show()

# Bar Plot

plt.figure(figsize=(6,4))

sns.barplot(

    data=df_bar,

    x="Category",

    y="Score",
```

```
        hue="Category",

        palette=["#4A90E2", "#7ED321"],

        legend=False

    )

plt.title("Mind vs. Relationship Scores")

plt.ylim(0, 40)

plt.tight_layout()

plt.show()

# Line Plot

plt.figure(figsize=(7,4))

plt.plot(

    mind_scores,

    marker="o",

    markersize=6,

    linewidth=2,

    label="Mind"

)

plt.plot(

    relationship_scores,

    marker="o",

    markersize=6,

    linewidth=2,

    label="Relationship"

)
```

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```
plt.title("Response Flow Pattern")

plt.xlabel("Response Order")

plt.ylabel("Response Intensity")

plt.grid(True, alpha=0.3)

plt.legend()

plt.tight_layout()

plt.show()
```

Listing 5.4: Python code illustrating the complete implementation of the *Noetica: Mind and Relationship Analyser*. The listing presents the integrated program structure, including user input handling, psychological scoring logic, interpretative analysis, and multiple visualisation outputs (pie chart, histogram, bar plot, and line plot). Inline comments are included throughout the code to clarify the purpose and functionality of each section and to demonstrate how theoretical concepts are translated into executable logic.

Overall, the inclusion of the fully commented code in this section demonstrates the structured and methodical approach adopted in the development of *Noetica*. By presenting the implementation alongside clear inline documentation, the project maintains transparency in program logic, supports ease of interpretation, and ensures alignment between the conceptual framework and the resulting executable output.

6. Outputs and Results

Noetica transforms user responses into meaningful insights, practical guidance, and visual representations. The output is structured to provide clarity, reflection, and actionable steps for personal and relational development. The results are presented in a logical sequence:

I. Analyses

Upon completion of the questionnaire, Noetica produces comprehensive analyses highlighting the user's current cognitive, emotional, and relational tendencies:

- a. **Mind Analysis:** Examines cognitive clarity, emotional regulation, self-reflection, impulse control, and mental resilience. It identifies areas of strength whilst highlighting recurring thought patterns, stress responses, and tendencies that may require attention. Users gain insight into how their mental processes shape daily decision-making, focus, and emotional well-being.
- b. **Relationship Analysis:** Evaluates empathy, communication skills, trust, boundary management, and relational consistency. It offers an understanding of how users connect with others, manage conflict, and sustain emotionally safe and supportive relationships. Patterns that may affect social interactions are highlighted for conscious improvement.
- c. **Integrative Insight:** Explores the interplay between mental clarity and relational competence. Users can observe how their inner state influences

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interactions with others and how relational experiences affect cognitive and emotional functioning. This holistic perspective emphasises the mutual influence of mind and relationships and supports integrated personal growth.

These analyses are interpretive, reflective, and actionable, providing users with a clear understanding of both strengths and opportunities for development.

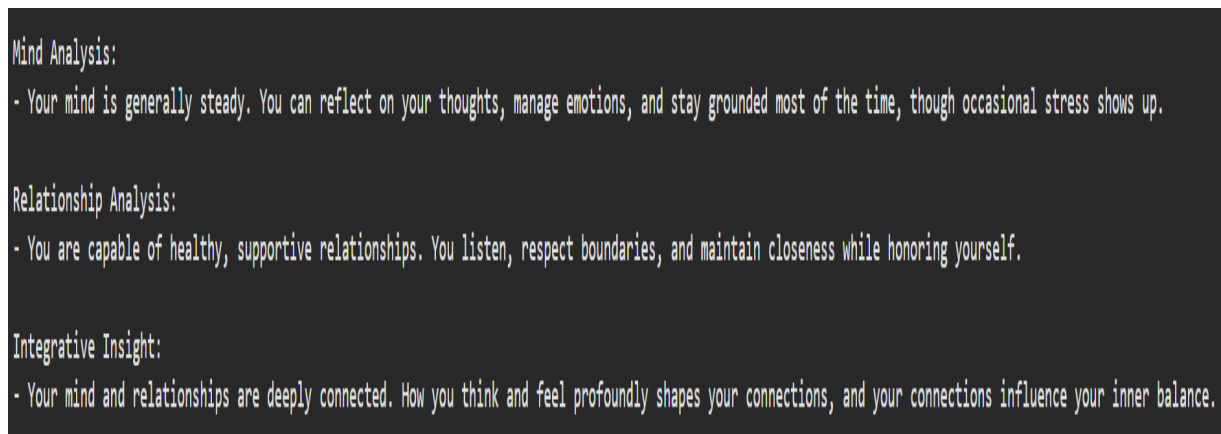


Figure 6.1: *Representative Output showing Mind Analysis, Relationship Analysis, and Integrative Insight.*

II. Suggestions

Following the analyses, Noetica offers tailored guidance to enhance self-awareness, emotional resilience, and relational health:

- a. **Mind Suggestions:** Practical steps for improving focus, reflective capacity, emotional regulation, and awareness of thought patterns. Recommendations include mindfulness practices, journaling, reflective pauses, and techniques for managing stress and impulsive reactions.
- b. **Relationship Suggestions:** Strategies to strengthen empathy, communication, trust, and healthy boundaries. Guidance includes active listening, expressing needs clearly, resolving conflicts constructively, and nurturing emotionally supportive and mutually respectful relationships.

These suggestions are designed for daily application, translating insights into meaningful behavioural improvements.

Mind Suggestions:

- Observe your thoughts and emotions regularly.
- Practice focusing on what matters most right now.
- Notice patterns without judgment and learn from them.
- Take time for reflective or creative activities.
- Check in with yourself during stressful moments.

Relationship Suggestions:

- Share feelings clearly and listen fully.
- Balance closeness with personal space.
- Work through small conflicts before they grow.
- Notice what makes relationships feel safe or tense.
- Appreciate the people who support you and be present for them.

***Figure 6.2:** Representative Output presenting Mind and Relationship Suggestions derived from the analysis.*

III. Seven-Day Practice Plan

Noetica generates a personalised seven-day reflective practice to consolidate self-awareness and conscious action:

- a. Encourages daily observation of thoughts, emotions, and relational behaviours.
- b. Promotes noticing patterns without judgement, suppression, or indulgence.
- c. Supports incremental growth in self-awareness, emotional regulation, and interpersonal skills.

This short-term practice bridges insight and action, assisting users in internalising lessons and gradually developing mindful cognitive and relational habits.

```
=====
7-DAY PRACTICE (RETAKE THIS TEST AFTER 7 DAYS)
=====
For the next 7 days:
- Choose ONE recurring habit or reaction.
- Watch it carefully every time it appears.
- Do not suppress it. Do not indulge it.

This strengthens conscious control.

After 7 days, take this test again and compare your scores.
Growth is measured by awareness, not perfection.
=====
```

Figure 6.3: Representative Output of the Seven-Day Practice Plan.

IV. Self-Insight Certificate

Upon conclusion of the assessment, Noetica produces a formal Self-Insight Certificate consolidating:

- a. Personal information (Name, Age Group, Gender, Date & Time)
- b. Scores for each domain (Mind, Relationship, Integrative Insight)
- c. Overall total score
- d. Interpretive conclusions, reflection quotes, and actionable guidance

The certificate provides a tangible record of self-reflection, allowing users to track progress over time and revisit insights for ongoing personal and relational development.

```
=====
SELF-INSIGHT CERTIFICATE
=====
Name           : Aarav
Age Group      : 18-25
Gender         : Male
Date & Time    : 23 December 2025, 04:10 PM
=====
Mind Score     : 26 / 40
Relationship Score : 28 / 40
Integrative Score : 2 / 2
Total Score    : 56 / 82

Conclusion:
- You have areas of strength and areas to grow. Paying attention to your mind and your relationships will help create balance and deeper fulfillment.

Reflection Quote:
"It is not what we have, but what we enjoy, that constitutes our abundance." - Pacal

Comment:
This certificate reflects current psychological patterns, not fixed traits. Awareness is the first step toward meaningful change.
=====
```

Figure 6.4: Representative Output illustrating the Self-Insight Certificate.

V. Visualisations

To enhance comprehension and engagement, Noetica offers optional visual outputs:

- Pie Charts:** Illustrate the proportional contribution of Mind and Relationship domains to the overall profile, providing a visual overview of balance between cognitive and relational competencies.
- Histograms:** Depict the distribution of responses (Yes, Maybe, No), highlighting consistency, uncertainty, and areas warranting further reflection.
- Bar Plots:** Compare domain scores to emphasise relative strengths and areas for development.
- Line Graphs:** Display sequential trends across the questionnaire, revealing patterns, fluctuations, and recurring behavioural tendencies.

These visualisations complement textual feedback, enabling users to engage with the data both analytically and intuitively.

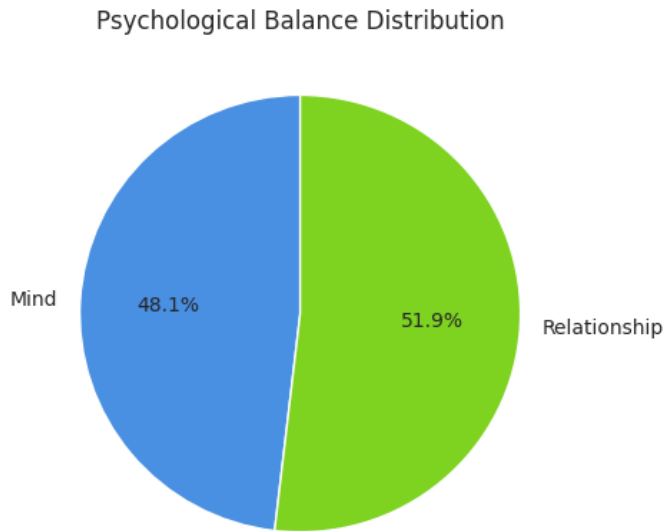


Figure 6.5.1: Representative Output of the Pie Chart

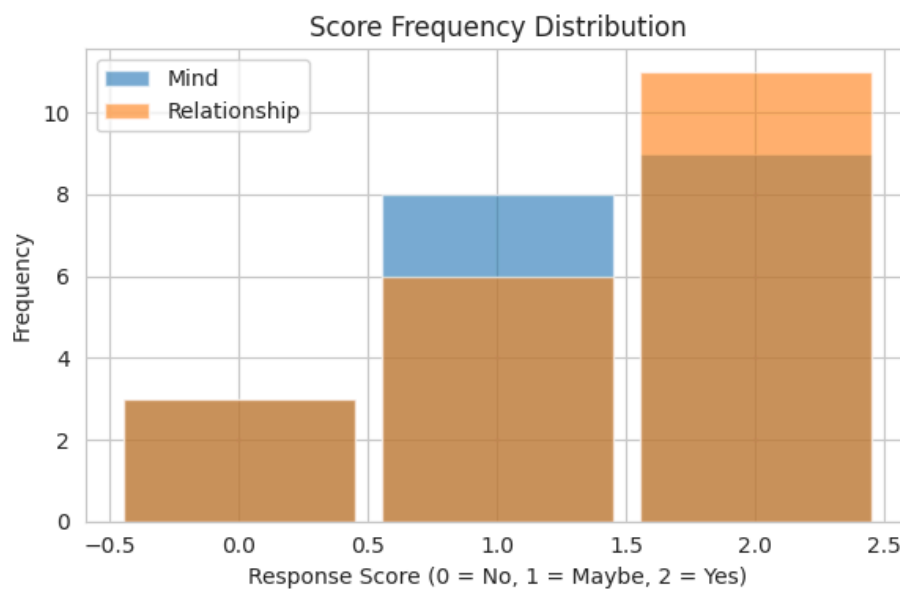


Figure 6.5.2: Representative Output of the Histogram

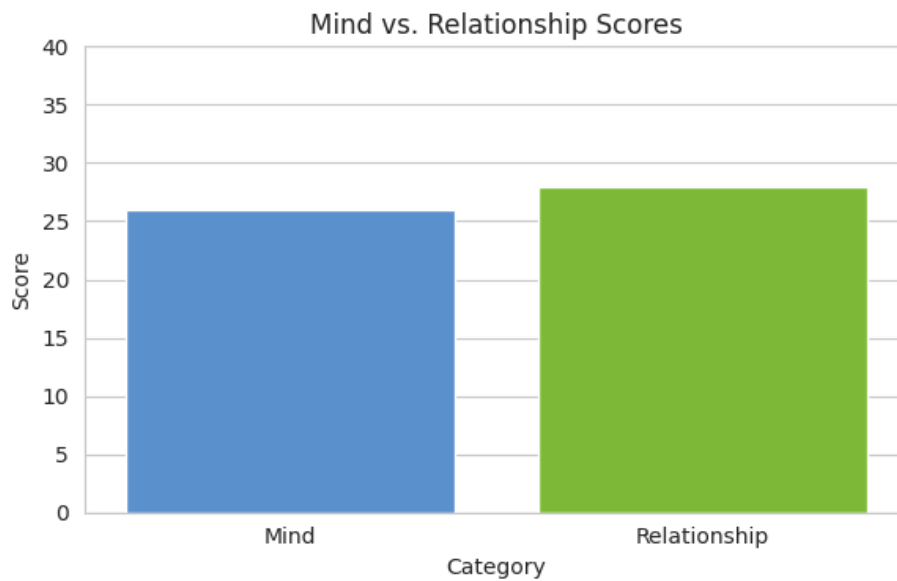


Figure 6.5.3: Representative Output of the Bar Plot

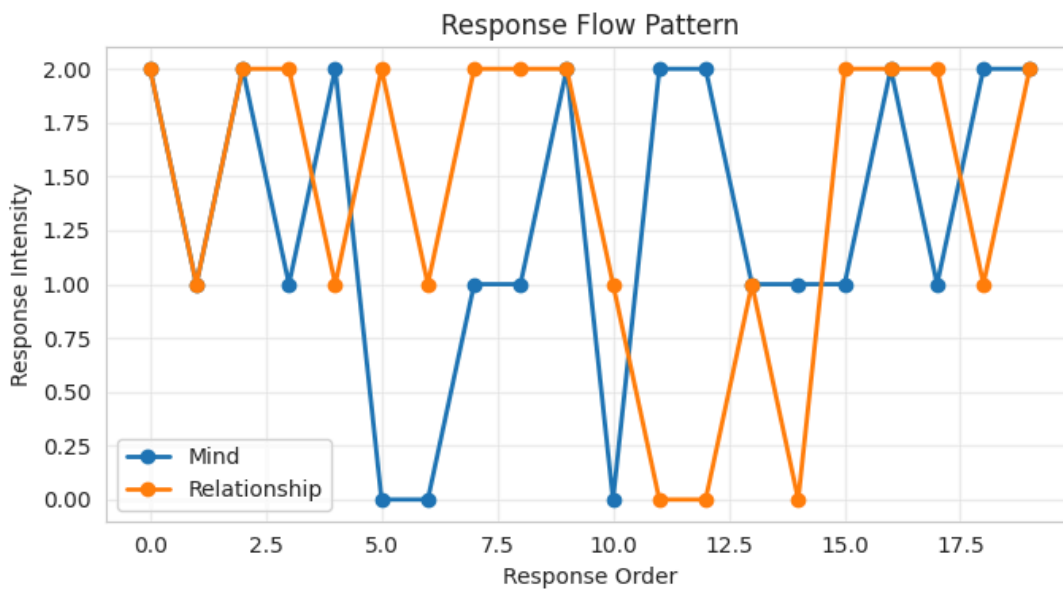


Figure 6.5.4: Representative Output of the Line Graph

7. Conclusion

Noetica: Mind and Relationship Analyser is a Python-based, structured, and interactive mini-project designed to bridge the gap between subjective self-reflection and objective assessment. By integrating quantitative scoring, conditional logic, qualitative feedback, and visualisation, the tool provides a multi-dimensional understanding of an individual's cognitive processes, emotional regulation, and relational dynamics.

The Mind domain enables users to gain insight into their internal cognitive and emotional landscape, offering clarity on self-awareness, impulse control, emotional stability, reflective capacity, and decision-making processes. By systematically highlighting strengths and areas requiring attention, the tool encourages deliberate engagement with mental patterns, fostering mindfulness, reflective practice, and improved resilience.

Similarly, the Relationship domain evaluates interpersonal competencies, including empathy, communication effectiveness, trustworthiness, and boundary maintenance. The analyses produced allow users to recognise both positive relational behaviours and areas of potential improvement. This facilitates the cultivation of healthier, more balanced, and mutually respectful interactions, promoting stronger and more fulfilling personal connections.

A distinctive feature of Noetica is the Integrative Insight domain, which examines the interaction between cognitive clarity and relational functioning. This domain demonstrates that mental processes and social interactions are interdependent. By understanding how internal states influence external relationships and vice versa, users are encouraged to adopt a holistic approach to personal development. The tool emphasises that improvements in one domain can positively influence the other, supporting comprehensive self-growth.

Noetica's outputs are both qualitative and quantitative, combining numerical scores with descriptive textual feedback, visual analytics, and a structured seven-day reflective plan. Textual feedback provides actionable guidance tailored

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to individual profiles. Visual outputs, including pie charts, histograms, bar plots, and line graphs, support intuitive interpretation of patterns and trends. The seven-day plan encourages daily reflection, careful monitoring of thoughts and behaviours, and incremental self-improvement. The generation of a formal self-insight certificate consolidates scores, analyses, and recommendations, providing a tangible record for future reflection and longitudinal tracking.

From a technical perspective, Noetica demonstrates the practical application of Python programming, data handling, conditional logic, and graphical visualisation within psychological self-assessment. Despite being a mini-project, it effectively translates introspection into structured, measurable, and actionable insights. The tool shows that even small-scale computational projects can contribute to meaningful self-understanding, cognitive and emotional growth, and relational improvement.

In conclusion, Noetica successfully provides a comprehensive, interactive, and evidence-informed framework for personal reflection and relational evaluation. By combining assessment, analysis, visualisation, and guided reflective practice, the tool empowers users to understand themselves and their social interactions more deeply. It supports structured reflection, encourages intentional behavioural change, and fosters continuous personal growth. Noetica demonstrates that structured self-assessment, even in a simplified computational format, offers substantial benefits in terms of insight, awareness, and actionable development, highlighting the value of integrating computational tools with psychological principles for holistic self-improvement.

8. References

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